

# Intro to Data Science

## Lab 2

### Linear Regression Basics

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## 1 Demonstrating Simple Linear Equation

File to use: Linear Equation Demo

Linear regression is based on finding a best-fitting line that is a linear equation that represents the relationship between two or more variables

A [basic intro to linear equations](#) for two variables: x and y

The simple Linear Equation worksheet demonstrates a basic linear equation for two variables: x and y, plotted as a scatter plot. Create a copy of it and play around with changing the key parameters of this equation (the value of b0 constant - the y-intercept and the value of b1 - the slope coefficient).

## 2 Demonstrating best fit line through Ordinary Least Squared Error (OLSE) method for linear regression

File to use: Best-Fit-Demo

The `Best-Fit-Demo-initial` worksheet shows the initial dataset and corresponding graph, whereas `Best-Fit-Demo-final` shows the final generated regression table

The dataset provides initial datapoints for the independent variable, Avg Temp, (values plotted on the X axis) and dependent variable, Sales (values plotted on the Y axis).

Select the Show Fit Line and Show Error Lines check boxes. Ensure the two other checkboxes are unchecked.

Try setting various values for the Y-intercept and Line Slope (using only the arrow buttons to increase / decrease the values), and view the linear equation line generated for these values. For each particular combination of values, you can see the sum of Squared error, which sums up the square of the distance from each point on the linear equation line (the regression model) to the actual datapoint.

The intention is to try and obtain a combination of values for the Y-intercept and Line Slope, so that the Sum of Squared Errors (SSE) is the smallest possible. Try playing around with setting various values for the Y-intercept and Line Slope to get an intuition for what we are trying to achieve.

Finally, click on the Show Trend Line (uses the underlying Excel Trend Line functionality). This is the regression model line whose values for the Y-intercept and Line Slope will result in the SMALLEST possible value of SSE for all possible combinations of values for the Y-intercept and Line Slope. This optimization is done by Excel's [TREND statistical function](#) in the background.

The trend line is a linear equation that is the simple linear regression model that best approximates the relationship between these two variables. The regression algorithm essentially creates this model when it is provided the table of dependent and independent variables as its input.

Try to set the values for the Y-intercept and Line Slope to be as close to these values as possible (using only the arrow buttons to increase / decrease the values).

Finally, to get predictions for this model (which will just simply be Y-values corresponding to the existing X-values using the regression model linear equation), simply enter the linear equation in the Prediction column and drag and drop.

In Cell E4, this should look like:

$=\$H\$2 + (\$H\$3 * C4)$

Drag and drop down to fill up the remaining cells in the Prediction column with the same formula.

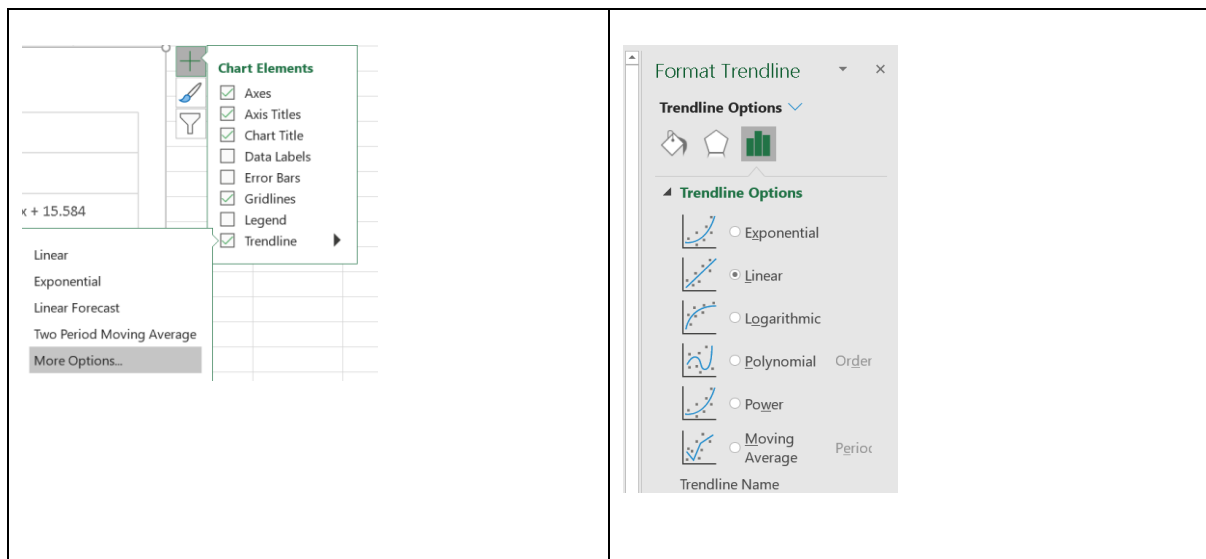
Now uncheck the Show Fit Line and click on the Show Forecast, which generates a linear graph line from the data in the newly populated Prediction column, and you will see this completely aligns with dotted Trend line we generated earlier, which is expected since both have nearly identical Y-Intercept and Line Slope values.

Finally, we can plug in new values for Avg Daily Temp box (the independent variable) and get predicted forecasts for the sales (the dependent variable) using the regression model linear equation.

This essentially demonstrates how a regression model linear equation is generated for a simple case of a single independent variable and dependent variable.

The Demo Different TrendLines worksheet demonstrates simulating a series of values for a dependent (y-axis) variable by adding / subtracting some random numbers from the results of a linear equation involving the independent (x-axis) variable. We can plot a scatter plot involving the new dependent (y-axis) variable and same independent (x-axis) variable, and generate a trendline for the scatter plot. Once again, the trend line will be the linear equation of the simple linear regression model that is the best approximate of the relationship between the dependent and independent variable.

Excel graphs offer multiple trendline options, here we are using the Linear Trendline for a linear regression model.



You can change the values of the upper limit and lower limit for the random number generated (which will also change every time the sheet is refreshed or saved) to view how different trend lines are generated for different distributions of random dependent variable values.

### 3 Demand forecasting example for Multiple Linear Regression: Estimating and interpreting model coefficients

Demand forecasting is a common predictive analytics use case for supply chain analytics. There many possible feature / independent variables that can be used here to train the ML model in order to predict demand (target / dependent variable) for a particular product. Typical variables that are used include historical sales statistics, seasonality dependent features, product features, location / market features, price, advertisement and promotion efforts, etc. These variables are typically determined based on the extensive real-world experience of the domain expert operating in the field.

We have a simple demand forecasting example here where we are going to train a model to predict the unit sales for a particular product for a given month. There are 3 independent variables to be used

- the price at which the product is sold (Price)
- the monthly amount that the company spends on advertising the product (AdExp)
- the monthly amount spent on promotions for the product (PromExp)

File to use: Basic-Sales-pt1

The `initial` worksheet shows the initial table, whereas `final-regression` shows the final generated regression table

You can make a copy from the initial worksheet and use that as a temporary worksheet and compare your final results with `final-regression`

We generate the regression model table using Excel's Analysis ToolPak:

Data ribbon tab -> select Data Analysis in the Analyze group

Highlight Regression from the Data Analysis dialog box, select ok

Complete the required info in the Regression Dialog box

Check Labels if you are going to select columns in the table with their headers included (recommended)

Select Input Y Range, then select the dependent / target / y variable (1 column)

Select Input X Range, then select the independent / feature / x variable (multiple columns)

Note: The X variable columns that you intend to use in your regression model must be stacked contiguously next to each other, if they are not you will have to rearrange the columns manually to make it so

**Note:** You can use shortcut Ctrl-Shift-Down arrow key to select an entire column / series of columns in a large table when the first couple of rows are already highlighted.

For output options, you can choose to generate the regression model in either the current worksheet in an empty location or a in a new worksheet. For here, we will generate in the current worksheet.

Select Output Range and select on a suitable empty cell that will become the upper left hand corner of the regression table to be generated.

Finally select Ok.

The required coefficients / parameters are in the 2<sup>nd</sup> column of the table

|                  | Coefficients |
|------------------|--------------|
| Intercept        | -25096.83292 |
| Price (\$)       | -5055.269866 |
| Adexp ('000\$)   | 648.6121403  |
| Promexp ('000\$) | 1802.610956  |

There is a variety of other important info generated in the table such as R-Squared, confidence intervals, etc that we will explore in later labs.

Interpretation for the current model:

**For every 1 unit increase in price (\$ 1), the unit sales drops by 5055 units**, with all other variables remaining at the same level. Conversely, for every 1 unit decrease in price, the unit sales increase by 5055 units.

**For every 1 unit increase in ad expenditure** (which in our example is \$1,000 as indicated in column header), **the unit sales increase by 648 units**, with all other variables remaining at the same level. This interpretation scales linearly, which means that if ad expenditure increases by \$10,000 (10 units), then we would expect the unit sales to increase by 6,486.12 units with all other variables remaining at the same level.

**For every 1 unit increase in promotion expenditure** (which in our example is \$1,000 as indicated in column header), the **unit sales increase by 1802** units, with all other variables remaining at the same level.

Lets assume that we are trying to maximize the unit sales.

In the simplest case, we locate the coefficient that results in the greatest unit of unit sales.

Based on the 3 coefficients we have obtained so far, we can conclude that decreasing the price by \$ 1 results in the greatest increase in unit sales (5055), compared to increasing ad expenditure by \$1,000 (increases sales by 648) or increasing promotion expenditure by \$1,000 (increases sales by 1802)

Note that in a real world situation, there may be practical considerations that come into play which prevent the application of this interpretation. For e.g. prices of products after often times fixed due to pricing strategies and operational concerns. So in that situation, you may have to resort to increasing promotion expenditure as the 2<sup>nd</sup> most effective way to increase overall sales.

When interpreting the beta zero coefficient, the estimate of the beta zero coefficient is the value of the y variable when all x variable values are zero. So, in this case, it implies that the value of unit sales would be -25096.83, when all the price is zero, the ad expenditure is zero, and the promotional expenditure is also zero. This technical interpretation of beta zero which does not have any managerial / operational relevance. This is because we are considering a situation where you're selling a product for free (price is 0), which would never make sense in the real world. Also negative unit sales also has no meaning when translated into real world situations.

In this situation, we say that the beta zero coefficient is a technical interpretation that is required merely to fit the model to data but has **no real life managerial / operational significance**. There could be some situations where the intercept also could have a managerial interpretation. And we'll look at those situations later on in some lab sessions.

## 4 Generating predictions from the model

File to use: Basic-Sales-pt2

The `initial` worksheet shows the initial table, whereas `final-prediction` shows the final table with generated predictions

We generate predictions from the model by constructing the regression linear equation from scratch and then substituting values for the various independent variables for the different scenarios that we wish to obtain a predicted value for.

This then helps us to determine which scenario maximizes (or minimizes) the dependent variable that we are interested in.

## 5 Exercise: Interpreting model coefficients and generating predictions

File to use: Demand-Forecast.xlsx

This exercise involves another sample dataset pertaining to demand forecasting, with the target variable as the order quantity but with a different set of feature / independent variables.

- a) days in the month that the store was closed
- b) the number of staff manning the store
- c) the price of the product
- d) the discount offered on that product (%)

The `initial` worksheet holds the empty dataset and a scenario for which the order quantity is to be predicted for a specific combination of values of the 4 feature variables.

Generate a regression model from this dataset, interpret the coefficients correctly, and generate the required prediction using these coefficients in the linear regression equation correctly. The final answer is in the `final-prediction` worksheet.

## 6 R-Squared for quantifying model prediction accuracy

File to use: `R-Squared Scenarios`

The `initial` worksheet demonstrates simulating a series of values for a dependent (y-axis) variable by adding / subtracting some random numbers from the results of a linear equation involving the independent (x-axis) variable.

Change the values for the upper and lower limit of the random number that is added to the Model Y values. The larger these values, the bigger the deviation of the simulated actual dependent values from the trend line that represents the regression model. This essentially represents a regression model that is less likely to generate accurate predictions.

Use these values for the upper / lower limit : 0, +- 20, +- 40, +- 60, +- 100

For each value that you use, generate a new regression table for that value (using the Y dependent and X independent value).

Notice that for the case of 0 for upper / lower limit, the simulated actual dependent values align perfectly with the trend line that represents the regression model. This represents a regression model that will always make perfectly accurate predictions, which in turn corresponds to a R-squared value of 1. Also notice the coefficients for the intercept and X correspond to the actual original linear equation ( **$y = 6x + 10$** )

For higher values of the upper / lower limit, the R-squared value of the regression model decreases correspondingly, reflecting a lower accuracy level of predictions generated from that particular regression model.

## 7 Adjusted-R Squared to drop meaningless variables (noise)

File to use: `Basic-Sales-pt3`

One of the key reasons for a low R-squared (which is due to high errors / residuals) is that important independent variables that contribute significantly to the dependent variable have been left out of the regression model / equation.

The easiest way to increase R-Squared is to **add extra relevant independent variables** to the regression model. However, there is a subtle problem here: R-Squared will either increase or remain the same when new independent variables are added to an existing model **REGARDLESS** of the relevance of these variables to the dependent variable.

This means that simply adding independent variable values that have **no contribution or effect** whatsoever to the overall regression model will still increase R-Squared. The end result of this is if that you add enough meaningless or non-significant independent variables, you will manage to artificially increase your R-Squared to a high enough value that gives a false / incorrect impression of an accurate regression model. This phenomenon is known as **overfitting in machine learning**.

To overcome this problem, we utilize Adjusted R-Squared.

Adjusted R-Squared will only increase when a newly added independent variable contributes meaningfully to the dependent variable. If it does not, the newly added variable can be considered as noise that can be dropped from the construction of the model.

The `initial` worksheet shows the initial table. Create a copy from this.

Generate 3 regression models from this table where the dependent variable is unit sales and the independent variable is

1. Price only
2. Price and AdExp
3. Price, AdExp and PromExp

The final result should be similar to that in `final-incremental-v1`. Here we can see that incrementally adding independent variables to the regression model / equation increases both R-squared and also adjusted R-Squared. Also, we can see incremental increase in R-Square due to the addition of each new variable is different for each variable. The bigger the incremental increase, the more significant that newly added variable is in the overall regression equation.

This provides another alternative way to determine the significance of each particular variable in influencing the dependent variable. We had seen earlier the simplest way to do this was to check the magnitude of the coefficient associated with that dependent variable (Section 3).

We can now repeat the previous process of generating 3 different regression models from incremental addition of the independent variables, but with a reversed sequence of adding these incremental variables:

1. PromExp
2. PromExp and AdExp
3. PromExp, AdExp and Price

The final result should be similar to that in `final-incremental-v2`.

By comparing the increment in R-Squared for these two processes, we can deduce the individual contribution of each independent variable to the overall R-Squared:

- a) Price (0.61 - 0.65)
- b) AdExp (0.04 - 0.06)
- c) PromExp (0.10 - 0.12)

We can see that Price here is the most significant independent variable by a far margin compared to the two other independent variables.

Next, start from the worksheet `initial-noise`.

This now has 2 new additional columns (Defective Units, Workers) that we can add as independent variables to the existing 3 independent variables that we already have.

Create another 2 new regression models involving adding in these 2 columns as independent variables incrementally. The final result should be as in `final-incremental-noise`.

The key point to note in this is that

- The value of R-Squared **increases** with each addition of these 2 new independent variables
- The value of Adjusted R-Squared **decreases** with each addition of these 2 new independent variables

This indicates that these 2 new independent variables are irrelevant and should not be included in the model. If they are included, they will affect the predictive accuracy of the model, even though R-squared is high. This is very useful when you have **many possible independent variables to add to the model and you are not sure which ones to incorporate**.

In addition to determining that these 2 new independent variables are irrelevant and can be safely left out from the model, we can also determine the relevance / importance ranking among the 3 independent variables that we have used by several different measures:

- a) Examining the p-value. The smaller the p-value, the more relevant / important the independent variable is
- b) Examining the coefficient magnitude. The larger the magnitude, the more relevant / important the independent variable is. Standardization should be performed first if necessary
- c) Examining the comparative change in R-Squared / Adjusted R-Squared when the variable is incrementally added / removed from the model. The larger the comparative change, the more relevant / important the independent variable is

## 8 Exercise: Identifying relevant independent / feature variables

File to use: `Predictive-Maintenance`

This exercise involves a sample dataset pertaining to predictive maintenance, another common predictive analytics use case for supply chain analytics. Typical feature / independent variables that are used to predict where a machine will fail or not include machine sensor-based IOT measurement data (such as temperature, vibration, noise levels, pressure, etc), usage data (cycles of operation, asset age, downtime, etc), maintenance history (MTBF, MTT, number of past failures, etc).

The `initial` worksheet shows the initial dataset with the target variable being the core temperature and we can evaluate machine failure to occur when the core temperature is over a certain threshold (for e.g. 100C). Currently we have 5 columns that could be potentially used as feature / independent variables to construct the linear regression model to predict the core temperature.

Using the approach we have examined in previous lab, determine which columns are strongly correlated with the target column so that they can be correctly used as independent variables to construct the linear regression model.



HINT: **At least one of these columns have little to no significance** on the target column of core temperature.

You may need to rearrange the columns to build up your regression model from scratch. Feel free to create additional worksheets within the workbook to facilitate your work.

Among the columns that you have chosen to include as independent variables, which one has the strongest contribution to the target variable of core temperature?

The final answer is shown in the `correct-variables` worksheet.

Notice that Downtime(hours) has a higher coefficient magnitude in comparison to Rotational Speed (RPM) but a lower p-value. This means that Downtime (hours) has a bigger impact on the target variable (based on coefficient magnitude), but a lower statistical significance relationship.

## 9 Using conversational AI tools to facilitate model development.

For datasets that can be fitted with simple regression models, we can use conversational AI tools such as ChatGPT, Gemini, Claude, Grok etc to facilitate this process.

We will typically get a small number of data points by performing random sampling from the initial large dataset that we wish to use for modelling, then convert it to a suitable format (CSV) for the AI tools to analyze.

You can do this with the `Basic-Sales.xlsx` and `Predictive-Maintenance.xlsx` files as a starting point. Remove unnecessary columns and rows and rename the column headers to a simpler form. The final sample CSV are also provided (`Basic-Sales.csv` and `Predictive-Maintenance.csv`)

### Basic-Sales.csv

The attached CSV contains a dataset that I want to perform predictive analytics on. The target variable to predict is Unit Sales, and the independent variables are Price, Adexp, Promexp. Analyze the dataset and tell me what is the best regression model for this dataset? Briefly explain your reasoning step-by-step and provide justification for your final answer.

### Predictive-Maintenance.csv

The attached CSV contains a dataset that I want to perform predictive analytics on. The target variable to predict is Core, and the potential independent variables to be considered for training the model are Humidity, DB, RPM, Air, Downtime. Analyze the dataset and tell me which are likely to be suitable independent variables to be included, and which ones should be dropped? What is the best regression model for this dataset? Briefly explain your reasoning step-by-step and provide justification for your final answer.